

Name:

Stat 1040, Fall 2000
Final Test, Thursday December 14, 12:00–1:50 pm

Show your work. The test is out of 100 points and you have 110 minutes.

1. The U.S. Bureau of Labor Statistics regularly collects information on the labor market. According to the bureau, workers employed in manufacturing industries earned an average \$577 per week in May 1999. Assume that this average is based on a simple random sample of 1600 workers selected from manufacturing industries and that the standard deviation of weekly earnings in this sample is \$80.

(a) (10 points) Find a 99% confidence interval for the average weekly earnings of all U.S. workers employed in manufacturing industries in May 1999.

(b) (4 points) We know that income does not follow the normal curve. Since this is the case, does the confidence interval you found in part a) make much sense? Explain.

2. (6 points) After Florida Secretary of State Katherine Harris certified the Florida results in the 2000 presidential election as a 537-vote Bush win, a Salt Lake City TV station held a call-in poll. The station asked its viewers to call in with their answers to the question:

“Do you believe Al Gore should stop trying to overturn legally certified votes in Florida and acknowledge that he has lost?”

4,237 people phoned in, and 3,482 said “yes”. The station then announced that “82% of Americans believe Al Gore should concede defeat.”

Identify at least three problems with this survey and the announced results.

3. (4 points) A political poll takes a simple random sample of 1000 registered voters from Palm Beach (around 400,000 registered voters) and an independent simple random sample of 1000 registered voters from Collier (around 80,000 registered voters). They plan to estimate the percentage of registered voters who think they voted for Al Gore. If everything else is similar for these two counties,
- (a) the accuracy in Palm Beach should be quite a bit lower than the accuracy in Collier.
 - (b) the accuracy in Palm Beach should be about the same as the accuracy in Collier.
 - (c) the accuracy in Palm Beach should be quite a bit higher than the accuracy in Collier.
- (Say which is true, no explanation is required.)
4. (4 points) Suppose that in a certain large Florida county, there are exactly 50% Bush voters and 50% Gore voters.
- A: getting exactly 100 Bush voters in a simple random sample of 200 voters from that county.
- B: getting exactly 50 Bush voters in a simple random sample of 100 voters from that county.
- Which of the following is true (no explanation is required):
- (a) A is more likely than B.
 - (b) A and B are equally likely.
 - (c) B is more likely than A.
5. (12 points) According to genetic theory, if we look at parents who are both heterozygous RH-positive, there is a 75% chance for their child to be RH-positive. I take a random sample of 400 children of such parents and I find that that 288 are RH-positive. Does the genetic theory seem to be appropriate for the population from which I sampled? Perform a statistical test. You must clearly state a null and alternative hypothesis, compute a test statistic and a P-value and state your conclusions.

6. “Do angry people have more heart disease?” In a 4-year study of a random sample of 8,474 people with normal blood pressure, their anger was measured using a specially designed test which measures how prone a person is to sudden anger. At the end of the study, physicians (who did not know the anger test results) determined whether the people had heart disease or not. The following results were obtained:

	Anger score			Total
	Low anger	Moderate anger	High anger	
Heart disease	53	110	27	190
No Heart disease	3057	4621	606	8284
Total	3110	4731	633	8474

- (a) (12 points) Is there evidence that anger and heart disease are related? You must clearly state a null and alternative hypothesis, compute a test statistic and a P-value and state your conclusions.

- (b) (3 points) This study is an example of

- a randomized, controlled, blind experiment.
- a controlled, blind experiment (not randomized).
- an observational study.

(Just say which one, no explanation is required).

- (c) (5 points) Suggest a possible confounding factor for this study and clearly explain why it might be a confounding factor.

- (d) (3 points) Assuming the data show that anger and heart disease *are* related, does this show that anger causes heart disease? Explain clearly.

7. For a group of 19 countries, researchers looked at the average wine alcohol consumption (average number of liters of wine alcohol per person per year) and the heart disease death rate (number of deaths from heart disease per 100,000 people per year) for each country. They found the following:

Wine Consumption (liters per year) : average = 3.03, SD = 2.4

Heart disease death rate (per 100,000) : average = 191.0, SD = 66.6, $r = -0.84$.

- (a) (6 points) Find the equation of the regression line for predicting a country's heart disease death rate from its average alcohol consumption.

- (b) (4 points) The correlation coefficient of $r = -0.84$ shows which (if any) of the following:

- i. T/F there is a strong negative association between the amount of wine alcohol a person drinks and their chances of dying from heart disease.
- ii. T/F this is an example of an ecological correlation so the correlation coefficient is artificially close to -1.
- iii. T/F countries where people drink more wine alcohol tend to have lower heart disease death rates, and vice versa.
- iv. T/F the more wine alcohol a person drinks, the lower their chances of developing heart disease.

For **EACH ONE** say whether it is true or false, no explanation is required. As always, we are asking which statements are correct interpretations of the correlation coefficient, not about what you believe to be true or untrue about alcohol consumption and heart disease.

8. I have 20 lightbulbs a large box. Unknown to me, 4 of these 20 bulbs are broken. I select 6 bulbs at random from these 20 bulbs to put in a chandelier. *Answer each of the following questions separately.*

- (a) (1 points) What is the chance that the first bulb works?

- (b) (3 points) What is the chance that the second bulb works?

- (c) (3 points) What is the chance that all 6 of the bulbs work?

9. (12 points) The spermicide nonoxynol-9 kills HIV in the test tube, so researchers hypothesized that it might be useful in protecting high-risk women from HIV. Other researchers argued that nonoxynol-9 might increase the risk because it is an irritant. In a study of 990 prostitutes, participants were randomly divided into two groups. The treatment group were given a nonoxynol-9 gel. The control group were given a similar-looking but inactive gel. When the study ended in May 2000, 67 of the 495 women in the treatment group were HIV-positive, and 44 of the 495 women in the control group were HIV-positive. Perform a 2-tailed test to decide whether the treatment and control groups were significantly different with respect to HIV. Clearly state your conclusions.
10. (3 points) The study in problem 9 is an example of
- (a) an observational study.
 - (b) a controlled, blind experiment (not randomized).
 - (c) a randomized, controlled, blind experiment.
 - (d) a simple random sample.
- (Just say which one, no explanation is required).
11. (5 points) The National Collegiate Athletic Association requires colleges to report the graduation rates of their athletes. At USU, 62% of all 157 student athletes who entered between Fall 1990 and Fall 1993 graduated within 6 years of starting. True or false, and explain:

An approximate 95% confidence interval for the percentage of all USU athletes who graduate within 6 years of starting runs from 54.3% to 69.7%.

Memory Aids

Please note that these are provided for your convenience, but it is your responsibility to know how and when to use them.

$$\text{rms error} = \sqrt{1 - r^2} \times SD_Y$$

$$\text{slope} = r \times \frac{SD_Y}{SD_X}$$

$$\text{intercept} = \text{ave}_Y - \text{slope} \times \text{ave}_X$$

$$\frac{n!}{k!(n-k)!} p^k (1-p)^{n-k}$$

$$\text{correction factor} = \sqrt{\frac{\text{number in the population} - \text{number in the sample}}{\text{number in the population} - 1}}$$

$$SD^+ = \sqrt{\frac{\text{number of draws}}{\text{number of draws} - 1}} \times SD$$

$$SD_{\text{box}} = \sqrt{\text{fraction of 0's} \times \text{fraction of 1's}}$$

$$EV_{\text{sum}} = \text{number of draws} \times \text{ave}_{\text{box}}$$

$$SE_{\text{sum}} = \sqrt{\text{number of draws}} \times SD_{\text{box}}$$

$$EV_{\text{ave}} = \text{ave}_{\text{box}}$$

$$SE_{\text{ave}} = \frac{SE_{\text{sum}}}{\text{number of draws}}$$

$$EV_{\%} = \% \text{ of 1's in the box}$$

$$SE_{\%} = \left(\frac{SE_{\text{sum}}}{\text{number of draws}} \right) \times 100\%$$

$$SE_{\text{diff}} = \sqrt{a^2 + b^2} \quad \text{where } a \text{ is the SE for the first quantity,}$$

b is the SE for the second quantity, and the two quantities are independent